

Spring 2025 Midterm Exam

Foundations of Data Science

Name: _____

Question:	1	2	3	4	5	6	7	8	9	10	11	12	Total
Points:	19	3	10	3	9	8	6	4	11	4	8	15	100

Instructions

- Make sure you have a copy of the Midterm Exam Reference Guide and the Table Reference.
- Select the correct response(s) or provide a written response depending on the question type. If a prompt asks you to write code, then you can provide your own code or use the provided template, if available. Try to provide your responses in the template blanks or boxed spaces provided. If you find that you need additional space, write your extended response(s) on one of the provided blank sheets of paper and number them, so we can connect your response to the question.
- You can assume the following code has been run, when you are writing your Python code:

```
from datascience import *
import numpy as np
import matplotlib+
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
```

- For multiple-choice questions (☐) , select only one answer.
- For multiple-answer questions (☐) , select all correct options. You can earn partial credit, but incorrect choices will lower your score. However, your score will never be negative.
- The open response questions will be graded as:
 - Full Points: The response is correct and may contain a very very small error.
 - Partial Points: A reasonable response was provided. The partial point value will depend on your response.
 - No Points: No reasonable attempt was provided.
- Once you are finished, turn in your exam and you are welcome to leave.

1. **Midterm Project.** You just completed the Midterm Project where you examined three health studies that were conducted over the last century. There was data about the leading causes of death over the last 120+ years, a study about how cholesterol levels correlate with heart disease, and the National Diet-Heart Study. In this section, you will answer several questions that come directly from that project and the material covered in it.

- (a) (3 points) Given the table `causes_of_death`, write code that will output the year that is associated with the maximum-adjusted death rate for cancer.

Sample Solution:

```
(
    causes_of_death.where('Cause', are.equal_to('Cancer'))
    .sort('Age Adjusted Death Rate', descending=True)
    .column('Year')
    .item(0)
)
```

- (b) (3 points) Given the table `causes_of_death`, write code that will output the average Age Adjusted Death Rate for Stroke during the 2010s decade (including years 2010 through 2019).

Sample Solution:

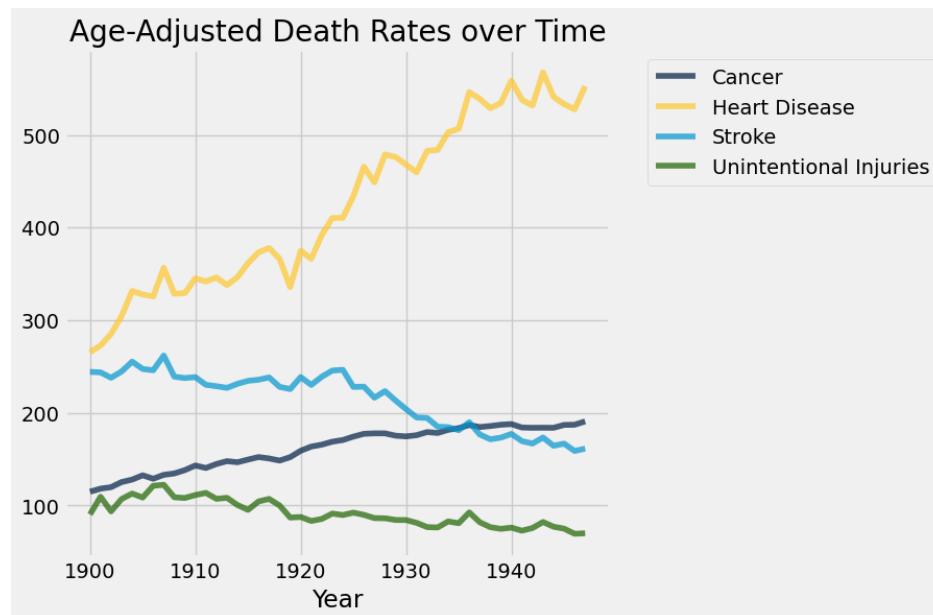
```
(
    np.average(
        causes_of_death.where('Year', are.between(2010, 2020))
        .where('Cause', are.equal_to('Stroke'))
        .column('Age Adjusted Death Rate')
    )
)
```

- (c) (3 points) Given the table `causes_for_plotting`, write code that will output the number of years between 1990 and 2023 where the Stroke age adjusted death rate was higher than the Cancer age adjusted death rate.

Sample Solution:

```
filtered_tbl = causes_for_plotting.where('Year', are.between(1990, 2024))
np.sum(
    filtered_tbl.column('Stroke')
    > filtered_tbl.column('Cancer')
)
```

- (d) (3 points) Given the table `causes_for_plotting` table, write code that will produce the following graphic showing the trend of causes of death from 1900 until 1947, inclusive. (You are not responsible for producing the title and there are no other customizations done to the graphic beyond using the `datascience` library table methods with `causes_for_plotting`.)



Sample Solution:

```
(
    causes_for_plotting.where('Year', are.between(0, 1948))
    .plot('Year')
)
```

- (e) (3 points) Given the table `framingham_by_id`, write code that will store the value of the number of subjects in the study who had an average cholesterol level (`'AVE TOTCHOL'`) less than 200 in the variable `low_chol_level`.

Sample Solution:

```
low_chol_level = (  
    framingham_by_id.where('AVE TOTCHOL',  
                           are.below(200))  
    .num_rows  
)  
low_chol_level
```

- (f) (2 points) In the Framingham Heart Study, you examined whether there was an association between cholesterol and development of heart disease. With a p-value of 0.0, can you conclude that higher levels of cholesterol caused heart disease? Why or why not?

Sample Solution: Although there were statistically significant results, this was an observational study, so we cannot conclude causation.

- (g) (2 points) In the National Diet-Heart Study at the Faribault institution, the subjects were randomly divided into two equal groups where half of the subjects were fed meals cooked with saturated fats (which was normal at the time) and half were fed meals cooked with polyunsaturated fats. Which of the following best describes the study design of the National Diet-Heart Study?

- ☐ An observational study
- ☐ A non-randomized, non-controlled experiment
- ☐ A randomized experiment
- ☐ A controlled experiment
- ☒ **A randomized, controlled experiment**

2. (3 points) **Hip Pain.** In an effort to investigate whether a new treatment is effective at reducing hip pain, researchers randomly sampled 100 patients from a medical group that used that new treatment, as well as other older treatments. They asked each patient whether or not they had received the new treatment or an older treatment, as well as whether or not they had experienced a reduction in pain. Overall, the results showed that those who received the new treatment observed a significant reduction in pain. Which of the following can be concluded from this passage? Select all that apply.

✓ **This is an observational study.**

☐ This an experiment.

☐ This is a randomized controlled experiment.

☐ The new treatment causes a reduction in pain.

✓ **There is a significant association between the new treatment and pain reduction.**

3. **Arrays.** Suppose we have executed the following lines of code.

```
ones = np.arange(5)
tens = np.arange(10, 50, 10)
hundreds = make_array(100, 200, 300, 400, 500)
```

What is the output associated with each line of code below? If there is an error, discuss the source of the error.

(a) (2 points) `ones + tens`

Sample Solution: This will produce an error. You can't add array of different lengths. More specifically: `ValueError: operands could not be broadcast together with shapes (5,) (4,)`

(b) (2 points) `ones + hundreds`

Sample Solution: Some version close to `array([100, 201, 302, 403, 504])`

(c) (2 points) `ones.item(3) * tens.item(1)`

Sample Solution: 60 or `3 * 20`

(d) (2 points) `ones * 3`

Sample Solution: `array([0, 3, 6, 9, 12])`

(e) (2 points) `np.diff(tens)`

Sample Solution: `array([10, 10, 10])`

4. (3 points) **Chocolates and Nutrition.** The two tables called `chocolates` and `nutrition` show some inventory data about a certain store's chocolates and the caloric data for each type of chocolate.

Here is a list of table method calls referencing these tables:

- A. `chocolates.group("Shape")`
- B. `chocolates.group("Shape", max)`
- C. `chocolates.group(["Shape", "Color"], max)`
- D. `chocolates.pivot("Color", "Shape", "Unit Price ($)", max)`
- E. `chocolates.join("Color", nutrition, "Type")`
- F. `chocolates.group(["Shape", "Color"])`

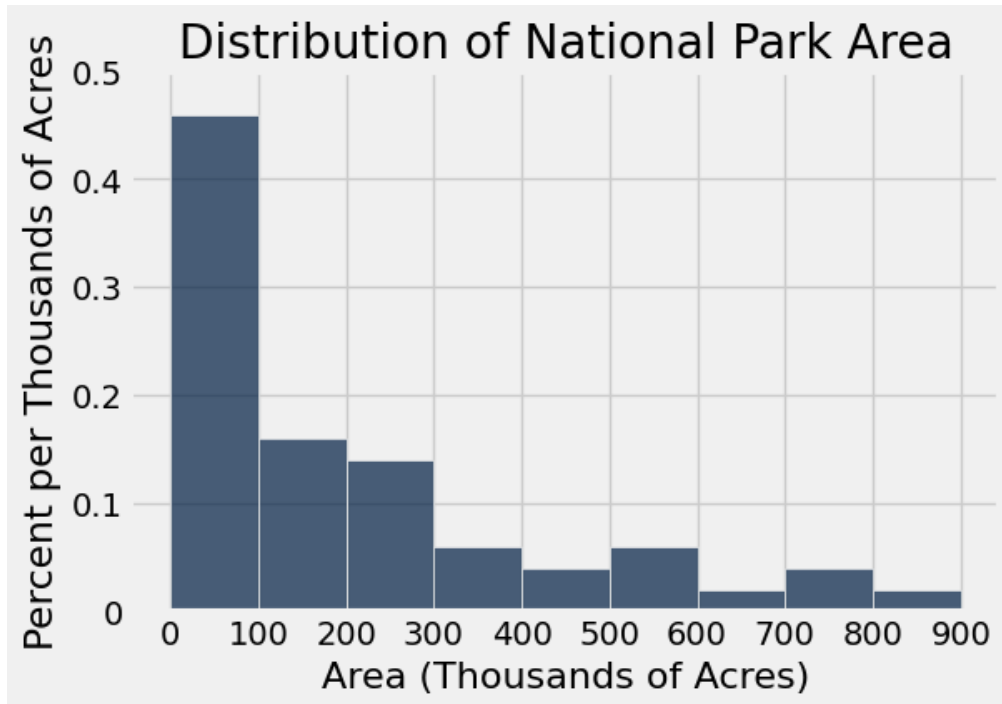
Write the letter of the table method call in the row that matches the description of the resulting table.

Table Method Letter	Columns in Table	# of Rows
B	Shape, Color max, Inventory max, Unit Price (\$) max	2
D	Shape, Dark, Milk, White	2
C	Shape, Color, Inventory max, Unit Price (\$) max	4
E	Color, Shape, Inventory, Unit Price (\$) max, Calories	6
A	Shape, count	2
F	Shape, Color, count	4

5. **Real Estate.** A real estate company has a data set of all of their buildings, with four attributes for each building: its size (in square feet), its type (residential or commercial), its ZIP code, and its estimated value if sold (sale price in dollars).

- (a) (2 points) The standard visualization to understand the distribution of building types is.
☒ **A bar chart** ☐ A line plot ☐ A scatter plot ☐ Overlaid histograms
- (b) (2 points) The standard visualization to check for an association between building size and building value is.
☐ A bar chart ☐ A line plot ☒ **A scatter plot** ☐ Overlaid histograms
- (c) (2 points) The standard visualization to compare the distributions of the estimated values (sale price in dollars) of the two types of buildings is.
☐ A bar chart ☐ A line plot ☐ A scatter plot ☒ **Overlaid histograms**
- (d) (3 points) Select all that are correct:
- ☐ The size attribute is a categorical variable.
 - ☒ **The type attribute is a categorical variable.**
 - ☐ The ZIP code attribute is a numerical variable.
 - ☒ **The estimated value attribute is a numerical variable.**

6. **National Parks.** We gathered a data set of US national parks, and plotted below a histogram of the size of these parks (in thousands of acres). All bars are 100 wide. The area of the visible bars sum to 100%.



Calculate each quantity described below or write Unknown if there is not enough information above to express the quantity as a single number (not a range). It's OK to write your answer as a Python expression or an unsimplified mathematical expression. You may need to estimate the height of bars visually; if so, make your best estimate.

- (a) (2 points) The percentage of parks that are less than 100 thousand acres in size.

Sample Solution: $0.46 \times 100 = 46\%$

- (b) (2 points) The percentage of parks that are between 200 to 400 thousand acres in size (specifically, the range $[200, 400)$, i.e., at least 200 thousand acres and less than 400 thousand acres).

Sample Solution: $(0.14 + 0.06) \times 100 = 20\%$

- (c) (2 points) The percentage of parks that are less than 150 thousand acres in size.

Sample Solution: Unknown. (We don't know the sizes of the parks in the [100, 200) bin; they could all be on the left half of the bin, or in the right half, or any mixture of those two.)

- (d) (2 points) The percentage of parks that are at least 1200 thousand acres in size.

Sample Solution: 0%. (It's to the right of all the bars.)

7. **Count Odd.** The code template for the function `count_odds` is below. The function takes in an array of numbers and returns the number of odd numbers in the array. Recall, the `%` operator returns the remainder if you divide by a certain number. For example, `11%3` would output 2.

- (a) (3 points) Use a combination of iteration and conditionals to complete the `count_odds` function below:

```
def count_odds(n_array):  
    num_odds = 0  
    for _____  
        if _____  
            _____  
    return _____
```

Sample Solution:

```
def count_odds(n_array):  
    num_odds = 0  
    for item in n_array:  
        if item % 2 == 1:  
            num_odds = num_odds + 1  
    return num_odds
```


- (b) (3 points) A researcher has collected data on various experiments, where each experiment generated multiple numerical results. The data is stored in a table called `results`. Using the `count_odds` function, write code to add a new column called `'odd_count'` to the table that contains the count of odd numbers in the `'results'` column for each experiment. After your transformation, `results` should contain an additional column showing the count of odd numbers for each experiment row.

Sample Solution:

```
results = results.with_column(  
    'odd_count',  
    results.apply(count_odds, 'results')  
)
```

8. (4 points) **Count Occurrences.** Which of the following functions correctly returns the number of occurrences of a specific value in a given array? For example,

```
count_arr_occurrences(make_array(0,1,0,5,1), 1)
```

should evaluate to 2 and

```
count_arr_occurrences(make_array("a", "b", "c"), "c")
```

should evaluate to 1. Select all that apply.

- ☐

```
def count_arr_occurrences(arr, value):  
    count = 0  
    for i in np.arange(value):  
        if arr.item__ == value:  
            count = count + 1  
    return count
```
- ☒

```
def count_arr_occurrences(arr, value):  
    count = 0  
    for x in arr:  
        if x == value:  
            count = count + 1  
    return count
```
- ☐

```
def count_arr_occurrences(arr, value):  
    return arr == value
```
- ☒

```
def count_arr_occurrences(arr, value):  
    return np.sum(arr == value)
```

9. **Smoothies.** Amy makes a smoothie every morning before going to school. To make her smoothie, she goes into the refrigerator and picks two pieces of fruit completely at random (first one, then the other). Suppose there are 2 pears, 5 mangos, 4 kiwis, and 1 peach in her refrigerator each morning when she goes to make her smoothie, and that each fruit is equally likely to be chosen.

For each part below, write an expression that evaluates to the probability described. You do not need to simplify any arithmetic.

- (a) (3 points) The probability that her next smoothie contains only kiwis.

Sample Solution: $(4 / 12) * (3 / 11)$

- (b) (3 points) The probability that her next smoothie has at least one mango.

Sample Solution: $1 - (7 / 12) * (6 / 11)$

- (c) (3 points) The probability that her next smoothie has a kiwi and a pear.

Sample Solution: $(4 / 12) * (2 / 11) + (2 / 12) * (4 / 11)$

- (d) (2 points) The probability that her next smoothie contains only peaches.

Sample Solution: $(1 / 12) * (0 / 11)$ or 0

10. **LinkedIn.** According to a recent survey, 28% of surveyed adults in the United States use LinkedIn. For the sake of this question, assume that the chance of a randomly sampled adult in the United States being a LinkedIn user is 28% (independently of all others).

- (a) (2 points) For which sample size below is there a higher chance that a random sample of that size will contain a percent of LinkedIn users of more than 50%?

☒ **20** ☐ 1,000

- (b) (2 points) According to the Law of Large Numbers (Law of Averages), with a smaller sample size the percentage of surveyed adults in that sample that use LinkedIn is more likely to be closer to 28% than a larger sample size.

☐ True ☒ **False**

11. **Wordle.** In the game of Wordle, a player guesses up to 6 words until they either correctly guess the secret word of the day or run out of guesses. Their guess count is either the number of guesses needed to guess the correct word (1 through 6) or X if all 6 guesses were incorrect. One thousand CCSF students played Wordle yesterday and the proportion of these students with each guess count is in the table below. We also created an array of these proportions called **students**.

1	2	3	4	5	6	X
0.0	0.17	0.33	0.27	0.20	0.02	0.01

```
students = make_array(0.0, 0.17, 0.33, 0.27, 0.20, 0.02, 0.01)
```

Wordle's creator, Josh Wardle, sent us the proportion of guess counts for all players who tried to guess yesterday's word and the table of proportions is below. The array called **everyone** contains all of these proportions.

1	2	3	4	5	6	X
0.0	0.09	0.25	0.32	0.28	0.03	0.03

```
everyone = make_array(0.0, 0.09, 0.25, 0.32, 0.28, 0.03, 0.03)
```

- (a) (2 points) What best describes the table for the students? Choose one.
- ☐ Probability Distribution
 - ☒ **Empirical Distribution**
- (b) (2 points) What is one way to simulate randomly selecting 1,000 individuals from the population of individuals that played Wordle yesterday? Choose one.
- ☐ `sample_proportions(1000, students)`
 - ☒ `sample_proportions(1000, everyone)`
 - ☐ `sample_proportions(1000, make_array('1', '2', '3', '4', '5', '6', 'X'))`
 - ☐ `sample_proportions(1000, make_array(1/7, 1/7, 1/7, 1/7, 1/7, 1/7, 1/7))`
- (c) (2 points) You would like to test whether the distribution of guess counts among the 1,000 students who played Wordle is different from the distribution provided by Josh Wardle. What test statistic could be used to run this hypothesis test?

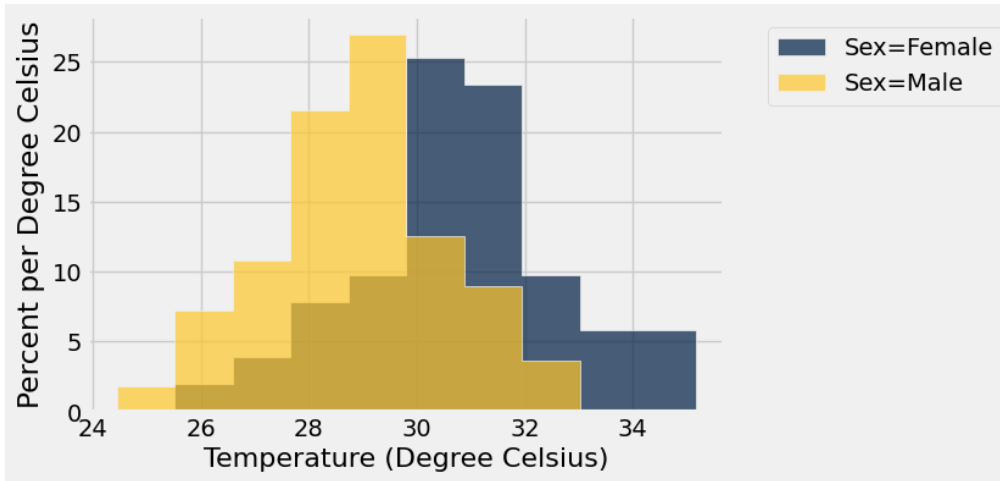
Sample Solution: The TVD can be used for this test.

- (d) (2 points) If you assume the distribution provided by Josh Wardle is similar for tomorrow, what is the chance that a randomly selected Wordle player will guess the word in less than 4 guesses?

Sample Solution: $0.00 + 0.09 + 0.25$

12. **Turtles.** Incubation temperature is an important factor in successful hatching of a baby turtle from an egg. Ellen read that the temperature at which an egg is incubated influences whether the turtle will be male or female. She loves turtles and is wondering whether this claim is true, or whether any observed differences might just be due to chance. She collects data on a random sample of 100 turtles and records both the incubation temperature (in Celsius) and the sex of the turtle that hatches in the table `turtles`.

Ellen decides to visualize her data before doing any inference. She creates the following overlaid histograms showing the distributions of the incubation temperatures for the female and male turtles.



Ellen would like to perform a test to see whether females in the population in general have higher incubation temperatures than males, or if the observed difference in distributions is due to chance.

- (a) (2 points) Which of the following is a valid null hypothesis for this test?
- ☐ Females in the sample in general have higher incubation temperatures than males.
 - ☐ There is no difference in incubation temperature between male and female turtles in the sample. Any difference we see in the population is due to chance.
 - ☐ Females in the population in general have higher higher incubation temperatures than males.
 - ☐ Males in the population in general have higher incubation temperatures than females.
 - ☒ **There is no difference in incubation temperature between male and female turtles in the population. Any difference we see in the sample is due to chance.**
- (b) (2 points) Which of the following is a valid alternative hypothesis for this test?
- ☐ Females in the sample in general have higher incubation temperatures than males.
 - ☐ There is no difference in incubation temperature between male and female turtles in the sample. Any difference we see in the population is due to chance.
 - ☒ **Females in the population in general have higher higher incubation temperatures than males.**
 - ☐ Males in the population in general have higher incubation temperatures than females.
 - ☐ There is no difference in incubation temperature between male and female turtles in the population. Any difference we see in the sample is due to chance.

- (c) (3 points) Ellen uses the difference between average incubation temperatures, defined by “female average minus male average” as her test statistic. In the following code cell, write a definition for the function `diff_of_means` that will take a table and a group label as arguments and returns the value of the test statistic. The table used as input (like `turtles`) has two columns – one of labels and one of numerical values.

```
def diff_of_means(tbl, group_col):
    means_table = _____
    means_array = _____
    return _____
```

Sample Solution:

```
def diff_of_means(tbl, group_col):
    means_table = tbl.group(group_col, np.mean)
    means_array = means_table.column(1)
    return means_array.item(0) - means_array.item(1)
```

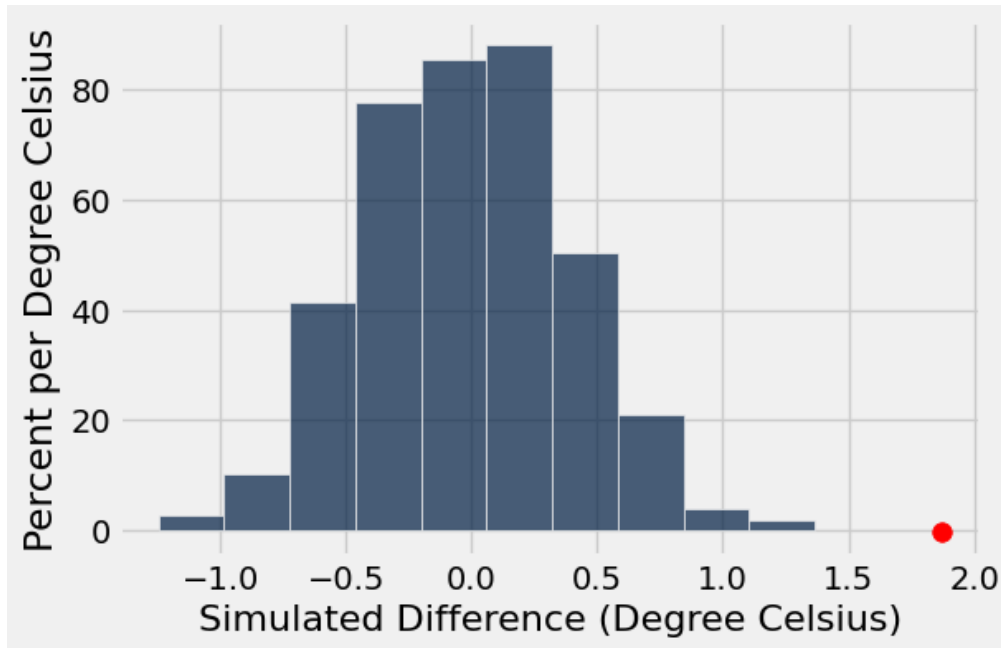
- (d) (3 points) Next, she simulates generating data many times under the null hypothesis. Complete the code below which will run a permutation of the data and calculate the test statistic 1,000 times, and store the results in an array called `diffs_array`.

```
diffs_array = _____
repetitions = 1000
for _____:
    resampled_table = turtles._____
    shuffled_labels = resampled_table.column('Sex')
    shuffled_table = turtles.with_column('Sex', shuffled_labels)
    shuffled_diff = _____(shuffled_table, 'Sex')
    diffs_array = _____
```

Sample Solution:

```
diffs_array = make_array()
repetitions = 1000
for i in np.arange(repetitions):
    resampled_table = turtles.sample(with_replacement = False)
    shuffled_labels = resampled_table.column('Sex')
    shuffled_table = turtles.with_column('Sex', shuffled_labels)
    shuffled_diff = diff_of_means(shuffled_table, 'Sex')
    diffs_array = np.append(diffs_array, shuffled_diff)
```

- (e) (3 points) Ellen ran `diff_of_means(turtles, 'Sex')` to get an observed value for the test statistic and assigned it to `observed_statistic`. The histogram below shows the distribution of the 1,000 simulated differences along with the observed difference, plotted with a dot.



Using `diffs_array` and `observed_statistic`, write code that will calculate the empirical p-value based on the simulation.

Sample Solution:

```
np.mean(diffs_array >= observed_statistic)
```

- (f) (2 points) Based off of the visualization, what can you conclude about Ellen's test? Assume a p-value cutoff of 5%.
- ☐ The data are consistent with the null hypothesis.
 - ☒ **The data are consistent with the alternative hypothesis.**
 - ☐ There is not enough evidence to make a conclusion.

Table Reference

Midterm Project

The `causes_of_death` table contains 496 rows and 3 columns. The `'Year'` column contains `int` values, the `'Cause'` column contains `str` values, and the `'Age Adjusted Death Rate'` column contains `float` values.

Year	Cause	Age Adjusted Death Rate
2023	Heart Disease	162.1
2023	Cancer	141.8
2023	Unintentional Injuries	62.3
2023	Stroke	39
2022	Heart Disease	167.2
... (491 rows omitted)		

The `causes_for_plotting` table contains 124 rows and 5 columns. The `'Year'` column contains `int` values and the `'Cancer'`, `'Heart Disease'`, `'Stroke'`, and `'Unintentional Injuries'` columns contain `float` values.

Year	Cancer	Heart Disease	Stroke	Unintentional Injuries
1900	114.8	265.4	244.2	90.3
1901	118.1	272.6	243.6	109.3
1902	119.7	285.2	237.8	93.6
1903	125.2	304.5	244.6	106.9
1904	127.9	331.5	255.2	112.8
... (119 rows omitted)				

The `framingham_by_id` table contains 4224 rows and 4 columns. The `'RANDID'`, `'AVE TOTCHOL'`, `'ANYCHD'`, and `'DEATH'` columns all contain `int` values..

RANDID	AVE TOTCHOL	ANYCHD	DEATH
2448	202	1	0
6238	249	0	0
9428	264	0	0
10552	228.5	0	1
11252	314	0	0
... (4219 rows omitted)			

Chocolates and Nutrition

The `chocolates` table contains 6 rows and 4 columns. The `'Color'` and `'Shape'` columns contains `str` values, the `'Inventory'` column contains `int` values, and the `'Unit Price ($)'` column contains `float` values.

Color	Shape	Inventory	Unit Price (\$)
Dark	Round	4	1.3
Milk	Rectangular	6	1.2
White	Rectangular	12	2
Dark	Round	7	1.75
Milk	Rectangular	9	1.4
Milk	Round	2	1

The `nutrition` table contains 4 rows and 2 columns. The `'Type'` column contains `str` values and the `'Calories'` column contains `int` values.

Type	Calories
Dark	120
Milk	130
White	115
Ruby	120

Turtles

The `turtles` table contains 100 rows and 2 columns. The `'Temperature'` column contains `float` values and the `'Sex'` column contains `str` values.

Temperature	Sex
30.1525	Male
30.7116	Female
30.778	Female
29.2344	Male
31.961	Female

... (95 rows omitted)